The Denial Spiral

How a Norm of Denial Prolongs Coercion,
and Why the Field Cannot Measure the Damage

Companion to The Deadly Race*: the obscurity class, its commons, and four mechanisms that censor its record*

Yuri da Silva Villas Boas · Working paper, v1, September 2026

Abstract

Its companion [11] derives a bar for coercion-resistant custody and shows that designs leaving a *feasible-but-unfinished* path manufacture an incentive to kill the holder. This paper takes up the other mechanism the same model yields, and the reason neither can be read off the record.

Denial credibility is a **common resource**: supplied by the population of holders, consumed by each holder who denies. A norm of concealment and coached denial depletes it, so that a denial carries no information and the rational continuation of an interrogation opening with one is *further coercion* rather than release. We name this the **Denial Spiral**, show it is the Obscurity Paradox run in reverse, and show that its cost is an *externality* — borne by whoever must be believed, including holders of designs that deplete nothing, and including people who hold no bitcoin at all.

We then classify what the industry ships against that mechanism — wallet-erase PINs, duress PINs, decoy passphrase wallets, and one vendor-documented instance of concealment marketed as deniability — and find the class fails under Kerckhoffs by construction, on the vendors' own documentation.

Finally we ask why a crisis this visible resists diagnosis, and give four compounding censoring mechanisms. Three remove the evidence: the exposure covariate is unmeasurable because obscurity does not advertise itself, the numerator is misclassified because a robbery whose victim is killed is recorded as a homicide, and the residue is held by bereaved families with no reason to speak. The fourth removes the *response*, and so survives publication: conceding is privately costly to whoever concedes while recalcitrance externalizes the cost — a commons problem one level up, with the field's own capacity to self-correct as the depleted resource. Coding of 351 registry incidents grounds the argument and bounds what it can claim.

The results are *derived*, not measured, and we say why no observation could settle them: every witness to an enacted case has an interest in the answer, so the instrument that would test the model is disabled by the mechanism the model describes.

Keywords: Bitcoin; self-custody; wrench (physical-coercion) attacks; obscurity; plausible deniability; duress code; Denial Spiral; Obscurity Paradox; Scheherazadian bargain; Bayesian adversary; praxeology; security economics; externalities; commons.

1 Introduction

The standing counsel to the Bitcoin holder who fears physical coercion has three parts: hold quietly, deny ownership if asked, and keep something to surrender if denial fails. Each is advice about what an attacker will *believe*. None of it has been assessed as what it is — a security mechanism whose strength depends entirely on the adversary's ignorance, which is the weakness *Reliance on Security Through Obscurity* catalogs [15].

This paper assesses it, and reaches a conclusion stronger than ineffectiveness. The counsel is *self-defeating in aggregate*, by a mechanism with the structure of a commons. Denial credibility is supplied by the population of holders and consumed by each holder who denies. Where denial

becomes the expected script, a denial stops carrying information; an interrogation that opens with one yields the attacker no exculpatory update; and its rational continuation is further coercion rather than release. Each holder's individually rational response to that environment — conceal harder, rehearse better — depletes the resource further, for everyone. We call this the **Denial Spiral** (§4), and its cost is an externality: it falls on whoever must be believed, including the holder of a design that depletes nothing, and including the person who holds no bitcoin at all.

Relation to the companion paper. [11] derives a bar for coercion-resistant custody — the *No-Gray-Area Criterion* — from the observation that a seizure leaving the attacker a *feasible-but-unfinished* path to the funds makes the holder a live competing racer, and so prices his removal. That is the *Deadly Race*: it governs the *terminal* move. The Spiral governs the *duration*. The two are companions rather than one being a case of the other: a design can be safe under one and guilty under the other, and the wallet-erase PIN of §3 is exactly such a design. This paper is self-contained — §2 restates what it borrows — but the formal development of the adversary, the criterion and the construction that clears it are in the companion and are not repeated.

Contributions.

1. The **Denial Spiral**: obscurity counsel as a depleting commons, with its cost falling outside the population that depletes it (§4).
2. A **classification of the shipped obscurity class** — wallet-erase and duress PINs, decoy passphrase wallets, and concealment marketed as deniability — against the Kerckhoffs premise, on vendors' own documentation (§3).
3. An account of **what kind of claim this is**: derived from purposive action rather than induced from observation, with an argument that no observation could settle it, because every witness to an enacted case has an interest in the answer (§5).
4. **Four censoring mechanisms** that keep the crisis undiagnosable, the fourth of which removes the response to the evidence and so survives the evidence being published (§6).
5. **Registry coding** of 351 incidents that grounds the stipulations and bounds what the argument may claim (§7).

2 What this paper borrows

Everything in this section is developed in the companion [11] and restated here only so that this paper can be read alone. Notation: $\mathcal{V}$ is the holder, $\mathcal{A}$ the coercer, $\mathcal{K}$ the *spend authority* — the set of states from which the stash can be spent, a single key being one case — and t the coercion-time budget.

Assumption 1 (The Wrench–Dolev–Yao (WDY) adversary)**.** $\mathcal{A}$ knows the protocol and the design landscape (Kerckhoffs), can seize every device co-located with $\mathcal{V}$, and can coerce $\mathcal{V}$ for a bounded time t. He is *priced*: he holds a value W for the stash and a cost c for a homicide, and he acts to maximize expected return. The construction is *a fortiori* — he is granted strictly more than a real coercer, so a bar cleared against him is cleared against the real one.

Assumption 2 (Non-Negative-Proof Principle, NNPP)**.** $\mathcal{V}$ cannot prove the non-existence of an undisclosed backup. Non-possession is a negative existential over everything in $\mathcal{V}$'s reach, the last site being whether $\mathcal{V}$ has forgotten or never memorized. $\mathcal{A}$ therefore cannot be brought to certainty by any disclosure $\mathcal{V}$ makes.

Two consequences of NNPP carry the weight here.

Corollary 1 (Disclosure)**.** *Cooperation cannot dispel the incentive: no act of disclosure closes the question, so compliance buys no guaranteed release [11, Cor. 1].*

Second, a denial is a claim $\mathcal{A}$ has both an incentive to disbelieve and no means of settling — which is the hinge of §4.

Definition 1 (Belief regimes, after [11, Def. 5])**.** NNPP fixes what $\mathcal{A}$ *cannot* settle. It does not fix what he does with that, and the companion separates two regimes, orthogonal to (W, c, t). **Dogmatic** $\mathrm{WDY}^{\mathsf{dog}}$ pins the posterior on "at least one undisclosed resumable copy remains" at 1 and never moves it — the limit of distrust, coherent precisely because NNPP holds, and the regime in which every bar in the companion is proved, since clearing it clears every attacker. **Bayesian** $\mathrm{WDY}^{\mathsf{Bay}}$ carries a prior over $\mathcal{V}$'s residual holdings and updates it on what the encounter discloses, then chooses among release, continued coercion and the terminal move by expected utility. By NNPP that posterior never reaches 0; it is simply not pinned at 1.

This paper works in $\mathrm{WDY}^{\mathsf{Bay}}$, and the division of labor follows from it. A bar must hold against the attacker who cannot be moved, so the Race is argued against $\mathrm{WDY}^{\mathsf{dog}}$. How long an encounter runs, and whether it ends in release, a long hold or a killing, is a question about an attacker whose assessment *does* move — which is why duration is this paper's subject and the outcome is the companion's. It is also why the Spiral has empirical purchase at all: what depletes is the informational value of a denial, and a denial can only lose value for an $\mathcal{A}$ who would otherwise have updated on it. Against $\mathrm{WDY}^{\mathsf{dog}}$ there is nothing for a denial to deplete, because nothing was ever going to be believed.

Definition 2 (Naive coercion resistance)**.** A protocol is **naively coercion-resistant** if it *may* resist a *sub-Kerckhoffs* attacker — one with the physical capabilities but lacking knowledge of the design — because denial, decoy or obscurity may work when the attacker does not know $\mathcal{V}$'s setup. Both hedges are load-bearing: the guarantee is contingent on the attacker's *ignorance*, not on any structural property, and so offers none against the full WDY adversary. This is the tier the whole obscurity class occupies.

Where the companion's stronger tiers are needed we name them in prose: a protocol is *weakly* resistant if a single encounter leaves a feasible-but-unfinished path and so a positive-pivotality state, and *strongly* resistant if no reachable state has positive pivotality [11, §3].

Lemma 1 (Deadly Race, after [11, Lem. 1])**.** *If, on release, $\mathcal{A}$ holds a feasible path to $\mathcal{K}$ while $\mathcal{V}$ retains one too, then $\mathcal{V}$ is* pivotal*: removing him raises $\mathcal{A}$'s win probability by $\pi > 0$, and the removal pays whenever $(1 - p_A)\, W > c$.*

Criterion 1 (No-Gray-Area, after [11, Crit. 1])**.** *A design must admit no such state: no encounter may leave a* gray *outcome, one in which $\mathcal{A}$ holds a state sufficient for $\mathcal{K}$ but has not completed the spend. Every encounter must resolve to $p_A = 0$ or $p_A = 1$.*

Finally, the companion's *Four Properties* are used here only to locate the obscurity class. They are **(1)** knowledge-based authentication, **(2)** individual custody, **(3)** non-obscurity — which *is* the Kerckhoffs premise — and **(4)** coercion resistance:

(1) knowledge-based authentication (deviceless);
(2) individual custody;
(3) non-obscurity;
(4) coercion resistance.

Clearing the bar while holding all four leaves two routes:

(R1) the *tacit* route, which removes the race itself [11, Cor. 3];
(R2) the **remote-delegate** route, which keeps a winning racer beyond the attacker's reach.

3 The obscurity class, and what vendors claim for it

The designs classified in the companion [11, §5] interpose a delay or a veto. A large and widely shipped class does something categorically different: it *erases*. An alternate PIN, when entered,

destroys the device's copy of the secret. Blockstream's Jade documents such a PIN under both of the names its help center uses for it — *duress* and *wallet-erase* — as an alternate PIN that "automatically delete[s] the encrypted wallet information from Jade if entered," presented as "especially useful in the event of a physical threat, as you can provide this PIN to an attacker." On entry the device reports only "Internal Error," so that the erasure reads as a malfunction [2, 3].

Grant the design the strength it genuinely has, and which the delay architectures of [11, §5] lack. The erasure is *instantaneous and covert*: $\mathcal{A}$ learns of it only once it is complete, so there is no interval in which he can intervene to prevent it. Nothing in what follows turns on $\mathcal{A}$ out-racing the mechanism, because he cannot.

The claim decomposes, and only one half is real. What the design offers $\mathcal{V}$ is not one mechanism but two, stacked and easily conflated:

(a) *the erasure* — a genuine, irreversible state change, which happens whatever $\mathcal{A}$ knows or believes; and

(b) *the belief* that nothing further remains to be had — manufactured by the "Internal Error" report and the return to setup, which are engineered to read as breakage rather than as a move. This is a *presentation* choice and not a property of erasure as such; Remark 2 records a vendor asked directly for it who declined.

Only a is Kerckhoffs-independent. The whole question is how much coercion resistance it carries on its own, and the answer, from the vendor's own instructions, is *none*.

The same documentation warns that afterwards "there is no way to regain access to your funds except by restoring using your recovery phrase," and directs the holder to "make sure you have a backup before proceeding" [3]. The feature therefore *presupposes* a surviving backup: a holder who follows the documentation has one, and a holder who does not has destroyed the stash rather than defended it. What erasure removes is thus not the secret but the *device path* to it — and a path that cost $\mathcal{A}$ nothing to lose, since a leaves $\mathcal{V}$ alive, present, and still holding everything needed to restore. In the ideal model a contributes exactly nothing to the floor, for the same reason a stage below the memory-hard chain contributes nothing to it (the shallow-stage null result in [11]): what is reachable without work is not a barrier.

All of the design's protective content therefore sits in b, which is an obscurity bet and nothing else. That places the design squarely in the naive tier (Definition 2): what protection it offers is contingent on $\mathcal{A}$'s ignorance of a documented feature, which is to say it forfeits non-obscurity (Property 3). It is worth being exact about which failure this is. The design does *not* fail the No-Gray-Area Criterion — an erasure destroys a path and names no second racer, so Lemma 1 does not reach it, and its row in the companion's design comparison [11, §5] is DR-safe and condemned regardless.

NNPP (Assumption 2) is what makes the obscurity bet unwinnable: $\mathcal{A}$ cannot verify that no further copy exists, and here he need not even reason his way to the suspicion, since the manual instructs the holder to keep a backup. He is therefore correct, by the design's own construction, to read the erasure as the closing of one route rather than the end of the matter. The route that remains runs through $\mathcal{V}$'s memory of a seed phrase, or of where a backup is kept: the erasure deletes the path that requires hurting no one and retains the path that does.

Remark 1 (The obscure self-destruct is the obscure scheme with no winning branch). Obscure defenses are usually compared by *how likely* $\mathcal{A}$ is to be ignorant. The sharper comparison is *how many* independent ignorances each one needs, and here the self-destruct is strictly the worse bet — worse, in particular, than the decoy it is commonly sold beside.

A decoy needs one premise: that $\mathcal{A}$ does not know decoys exist. Grant him that ignorance and the decoy *terminates the encounter* — he leaves with a plausible balance and a story that closes.

A self-destruct needs two, and they are independent. It needs $\mathcal{A}$ ignorant of the feature (sub-Kerckhoffs), so that he reads the failure as breakage rather than as a deliberate move; and it needs him ignorant of NNPP's consequence (sub-NNPP), so that he does not reason to an undisclosed

backup. The second is not implied by the first, and this is the design's central weakness: *even total success at the first leaves the encounter open.* "This device is broken" does not entail "this person has no bitcoin"; it entails that the instrument for reaching it is broken, whose natural sequel is the question *where is your backup?* — addressed to a $\mathcal{V}$ who is still present and still knows.

So the self-destruct fails against an informed $\mathcal{A}$ for the ordinary Kerckhoffs reason, and against an ignorant one it fails anyway, because the belief it manufactures does not discharge the encounter. It is the one obscure scheme with no branch on which it wins.

We did not have to derive that regress: a vendor documents it. Coldcard ships trick PINs whose actions include a duress wallet — "a personal safety feature… If you must reveal a PIN under duress, give the duress PIN instead of your Main PIN" — alongside *Brick Self* and *Wipe Seed* [10]. Its firmware documentation then addresses the case where the decoy is not believed, and its advice is to escalate to destruction: "you may be asked what the actual duress PIN is, while under duress. We suggest providing the 'brickme' PIN in that case" [9]. That is the manufacturer contemplating an $\mathcal{A}$ who knows decoys exist, and recommending, as the answer, an irreversible erasure performed with $\mathcal{V}$ still in the room and still able to be asked for a backup — the ignorance-count argument above, arrived at independently and published as guidance. The same document instructs the holder to "triple-check your backups are correct" before enabling that PIN, so the backup presupposition of §3 is not peculiar to one vendor either.

The same page goes further, in two admissions worth quoting because a vendor rarely makes them. The first concedes that the decoy is distinguishable to the $\mathcal{A}$ who bothers: "if you are somehow facing an attacker who is willing to verify he has the real main PIN, it's possible that careful analysis of system responses will imply he's working with the duress PIN." The remedy offered is not a repair but a request — "if you discover any sequences that reveal this easily, please tell us and we'll see if we can cover them up better" [9]. That is obscurity stated as a maintenance program, and it is the clearest statement we have found of what Property 3 rules out. What is being defended is not that the decoy is indistinguishable, but that no one has yet published how to distinguish it; the defense is administered by patching the disclosures as they arrive. A scheme whose security is the current state of the disclosure backlog is exactly the scheme whose guarantee $\mathcal{V}$ cannot invoke in the room, because $\mathcal{V}$ cannot know which sequences $\mathcal{A}$ has read.

The second admission is the denial move itself, offered where bricking is unpalatable: "alternatively, you can say you set the duress PIN once, but have since forgotten it." The escalation to destruction and this are the document's two exits, and both are the naive tier. The claim *I have forgotten it* is a proposition about $\mathcal{V}$'s own knowledge, and NNPP (Assumption 2) is precisely the observation that $\mathcal{V}$ cannot close it: there is no move that converts a true *I cannot* into an $\mathcal{A}$-verifiable one, and by Definition 2 the scheme's fate on that branch is decided by whether he believes it. The vendor has enumerated the branches and found, on each, a bluff.

That the line is published guidance sharpens rather than softens it, in the way Remark 5 describes for the coached decoy. A script printed in a manufacturer's manual is a script $\mathcal{A}$ can read as easily as $\mathcal{V}$; the sentence that teaches the excuse also teaches that the excuse is taught. Its credibility declines in proportion to its circulation, and the holder who most needs it inherits the discount without ever having been told it exists.

Remark 2 (The class, and what each vendor actually claims for it). Three vendors ship the pair, which is what makes this a design consensus rather than one firm's misstep; but they do not make the same claims, and the difference is worth recording precisely. Blockstream markets the *erasure* for physical threat while claiming nothing of the kind for its passphrase wallets. Coldcard markets *both* halves, describing the duress wallet as a safety feature and the brick PIN as the fallback when it fails. Trezor is the mirror image of Blockstream: its passphrase wallets are sold on deniability, while its wipe code — which the vendor itself calls a "self-destruct" PIN, and which erases the recovery seed with it — is documented without any mention of duress, coercion or physical threat [17]. That vendor was moreover asked, in January 2022, to make the wipe *concealable*: a user-defined message on entry, so that an erasure could be presented as a hardware

fault rather than announcing itself. The request was closed as *not planned*, and the device still reports the wipe plainly [5]. Component b is therefore not intrinsic to the self-destruct class at all: it is a presentation one vendor builds and another, asked for it directly, refused. What §3 analyzes under b is specific to the "Internal Error" design, and a wipe that announces itself carries a alone — which, by the argument above, is no coercion resistance at all, and is not sold as any. Two of the three therefore decline to make the coercion claim on one of their two mechanisms. We note it because the criterion is not advanced against vendors, and a vendor that ships a mechanism without claiming it defeats coercion has not made the error this section is about.

Remark 3 (Two bluffs the erasure invites). An $\mathcal{A}$ who knows the design (Assumption 1) need not walk into the erasure at all. Two alternatives stand open to him, each a wager rather than a certainty:

1. *Take the device hostage up front.* Rather than enter any PIN, seize the device and hold it as a hostage token (the hostage-token property in [11], the hostage-token remark in [11]). An erase PIN is irrelevant to a coercer who never authenticates; and by the backup-probe remark in [11] the terms on which $\mathcal{V}$ ransoms the device are themselves a probe for the very backups $\mathcal{V}$ cannot disprove.
2. *Coerce against deployment.* Brutalize $\mathcal{V}$ to deter the PIN's use in the first place, wagering that a holder under sufficient duress will not spend the one irreversible move available to them.

Neither wager is sure to pay, and that is precisely the point: both are bluff games, and the holder who has bet on an erasure has entered one without holding the means to end it. A $\mathcal{V}$ who erases and then *truthfully* denies retaining any backup arrives at the position the registry already records — the streamer whose denial was simply disbelieved, and who was tortured for hours regardless [14] (§7).

The "Internal Error" disguise inherits the same objection from a different direction: it is a bet on $\mathcal{A}$'s ignorance of a consumer product, and it degrades exactly as that product's own public documentation becomes known (the Four-Properties development in [11]). The class therefore lands in the **naive** tier of the time-gating tiers in [11] precisely as that definition describes it — an obscure self-destruct, safe only against a sub-Kerckhoffs attacker — and it is worth keeping the self-destruct distinct from the *cancel* key named alongside it there. A cancel reverses a pending action and thereby names a second racer; a self-destruct destroys a path and names none. Only the former is subject to Lemma 1, which is why the two fail differently despite sharing a tier. The perceptual-oracle-variants remark of [11] has already disposed of the tier as a whole, by the dilemma that an $\mathcal{A}$ whose ignorance can be assumed makes the elaborate mechanism unnecessary, and one whose ignorance cannot makes it ineffective; the present section is that dilemma instantiated in a shipping product.

The companion decoy, classified separately. The same device ships a second and quite different obscure mechanism, and the two should not be run together. A BIP-39 passphrase "is added to your Jade's recovery phrase or SeedQR and is used to create your wallet"; "any number of passphrases can be entered to change the wallet you will log into"; and the device "will not store your passphrase" [4]. Distinct passphrases therefore yield distinct wallets from one recovery phrase, and the familiar community practice is to keep a low-balance wallet at the empty passphrase to be surrendered under coercion.

Two observations. First, and to the vendor's credit, *Blockstream does not market it this way*: the passphrase documentation claims no deniability, decoy or duress use, in pointed contrast to the erase-PIN pages, which name physical threat explicitly. The duress framing here is the community's, not the manufacturer's, and the distinction is worth preserving in any account of who claimed what.

Second, the mechanism is the plausible-deniability layer already treated in the Four-Properties development in [11] and dismissed for the same Kerckhoffs reason: it rests on $\mathcal{A}$'s ignorance, and an $\mathcal{A}$ who knows that passphrase wallets exist simply persists past the surrendered one, with NNPP guaranteeing $\mathcal{V}$ cannot close the question. It is not, however, the *same* failure as the

self-destruct's, and by Remark 1 it is the less bad of the two: the decoy at least has a branch on which it wins, since an $\mathcal{A}$ who believes it has been paid and can leave. The self-destruct has none. That a single product ships both, marketing only the weaker one for physical threat, is the clearest illustration in this section of how little the design space has been mapped against the Criterion.

Remark 4 (A vendor-documented instance of the naive tier). The decoy's duress framing was the community's, not the manufacturer's. One vendor does make the claim in its own documentation. Sparrow's FAQ, answering "How can I choose where my wallet is saved?", documents a data-directory flag and adds that "[t]his feature allows you to store all Sparrow data on removable media making for more plausible deniability" [20].

The feature does not guard the asset. Sparrow's own best-practice page has the seed words backed up separately from the device, "ideally in a different location" [19] — so the concealed directory is neither necessary nor sufficient for $\mathcal{K}$. A Kerckhoffs $\mathcal{A}$ coerces the seed backup, loads it into a wallet of his own, and never touches the concealed medium at all. What is left for the medium to do is carry the holder's belief that, having surrendered what was found, they may now say there is nothing more. The decoy at least stands between $\mathcal{A}$ and the coins; this stands nowhere near them.

That is Definition 2 exactly — a guarantee contingent on $\mathcal{A}$'s ignorance, here of a *documented* feature — and the continuation is the one the registry records. A denial offered into the prevailing norm is pre-discredited (Corollary 1), so the rational continuation of an interrogation that opens with one is more coercion, and the truthful denier is brutalized alongside the lying one (§7). The feature adds nothing but the confidence to make the denial.

Remark 5 (The decoy is sold, and coached for credibility). The decoy is not merely a practice the community tolerates; it is a product and a curriculum. A Spanish-language self-custody consultancy lists, among paid service tiers, "*Passphrase (tu bóveda señuelo)* — Billetera señuelo + billetera real. Protección ante coacción/coerción" — a decoy wallet marketed, in as many words, as protection against coercion [21]. Published guidance goes further and coaches the performance: one guide has the holder keep a small balance on the bare seed as "what you'd hand over under physical coercion", and adds that "for this to work, the decoy balance must be *believable* — enough to look like a real wallet, not enough to devastate you if lost" [1]. The counsel is also delivered live: one of us appeared as guest on a self-custody program [16] in which the host worked through a decoy-construction walkthrough over the guest's contrary advice.

That last instruction is the naive tier stated as a training objective. A decoy need be believable only to an $\mathcal{A}$ who might disbelieve it, and the counsel is silent on what happens when he does — which, by Definition 2, is the whole question. A better-rehearsed lie raises the chance $\mathcal{A}$ takes the bait at the margin; it does not create a branch on which the scheme wins against an $\mathcal{A}$ who knows decoys exist, and NNPP (Assumption 2) guarantees $\mathcal{V}$ cannot close that question however well the performance goes. What is sold as a mechanism is a wager on the attacker's ignorance, priced and taught as though it were a control.

The counsel is also self-defeating in aggregate. Publicized training raises the *prior* that a fluent denial was rehearsed: once $\mathcal{A}$ knows credibility is taught and sold, fluency stops being evidence of truth and becomes evidence of coaching. The remedy degrades the thing it sells — and not only for its customers. A holder who never trained now denies into an $\mathcal{A}$ who discounts fluent denial generally, so the cost falls on people who bought nothing. This is the Obscurity Paradox (Remark 6) run backwards: there, a public non-obscure design makes $\mathcal{V}$'s claims *more* plausible; here, a public obscure training makes them *less*.

Remark 6 (The Obscurity Paradox — non-obscure designs strengthen obscure ones). The Time-Gated Perceptual Oracle of the companion [11, §4] enters neither bluff. There is nothing to erase, the deep secret having never been on the device (the floor theorem in [11]), and so nothing whose non-deployment can be coerced. But the comparison yields more than the absence of a failure mode, and the surplus runs in a direction the obscurity argument would not predict.

An obscure defense asks $\mathcal{A}$ to believe a *claim* — that this is the only wallet, that nothing further remains — against both an incentive to disbelieve it and, by NNPP, no means of settling it. A non-obscure defense changes what making that claim costs, and it does so by the mechanism set out in the adversary-education remark in [11]: an $\mathcal{A}$ who credits that an effective mechanism is in place is crediting that lying is not $\mathcal{V}$'s only available tool, and a holder with effective alternatives has correspondingly less reason to resort to one.

The ordering this inverts is worth stating plainly, and we name it **the Obscurity Paradox**: *a publicly specified, Kerckhoffs-clean design improves the standing of precisely those obscure denials it was built to make unnecessary.* Obscurity remains unsound as a primary defense, for the reasons the Four-Properties development in [11] gives. It is not, however, indifferent which design it accompanies — a denial made from within a mechanism that removes the need to lie is a stronger denial than the same words spoken from within one that requires it. The same asymmetry has an epistemic cost, taken up in §6 (Remark 10).

4 The Denial Spiral

Remark 7 (The Denial Spiral)*.* The paper has now met two distinct ways a custody practice can worsen its holder's position — the Deadly Race, borrowed in §2 and derived in the companion, and one that still needs a name.

Denial credibility is a *common resource*: it is supplied by the population of holders and consumed by each holder who denies. Under the adversary of Assumption 1, a norm of concealment and coached denial depletes it. Where denial is the expected script a denial carries no information, so an interrogation that opens with one yields $\mathcal{A}$ no exculpatory update and its rational continuation is further coercion rather than release (Corollary 1). The individual holder's rational response to that environment — conceal harder, rehearse better — depletes the resource further, for everyone, including the truthful denier who bought nothing (Remark 5).

We call this the **Denial Spiral**. It is the Obscurity Paradox (Remark 6) run in reverse, and it is the companion of the Deadly Race rather than a special case of it: the Race prices the *terminal* move, the Spiral prices the *duration* — the abductions, long holds and torture the registry codes (§7). A design can be innocent of one and guilty of the other, which is why they are named apart.

Corollary 2 (The Spiral is not a sixth axis either)**.** *Whether a design* depletes *the denial commons is equivalent to its failing Property 3. A design whose operation requires the holder to deny — a decoy, a concealed erasure — adds a scripted denial to the population and lowers what any denial is worth; a design that does not, and whose specification is public, does the reverse, since a denial made from within it is consistent with an ordinary setup rather than being the script (Remark 6). Depletion therefore tracks non-obscurity exactly, and a Denial Spiral column would reproduce that one as surely as a no-hostage column would reproduce Property 1.*

Exposure *is a different matter, and it is not a property of any design. The Spiral's cost is ambient: it falls on whoever must be believed, including the holder of a design that depletes nothing, and including the person who, on his own account, held no bitcoin at all — the RATIRL case (§7), and the reason the externality is worth naming. The qualifier is owed: §7 records that the denial may itself have been defensive, and the externality argument is the branch on which it was true. No column can carry that, because no column has a row for the people it lands on.*

4.1 "Take me with you" — the Scheherazadian bargain

What follows is a **conjecture**, not a result. Nothing else in this paper rests on it, there is no empirical evidence confirming any particular instance of the chain of events below, and it is set down here rather than withheld because the silence around it is over-determined by this paper's own §6: a headline records that a holder was held for eleven hours, never whether those hours went on interrogation or on waiting for a path to clear.

Remark 8 (The Scheherazadian bargain). At any point in an encounter on a Deadly-Race-vulnerable setup, $\mathcal{A}$, $\mathcal{V}$ or both may work out Lemma 1. A WDY^{Bay} attacker (Definition 1) who has updated his way to being satisfied that he holds everything may *still* not release, because a released holder with a feasible path is a competitor; detention buys the same assurance as the terminal move at a fraction of the judicial disutility. Gray-area vulnerability therefore prices **custody as collateral on $\mathcal{V}$'s own abstention**. The conjecture is that $\mathcal{V}$, having run the same arithmetic, will *volunteer* for it: *take me with you, and you will know I am not racing you.*

We call it the **Scheherazadian bargain**. In the *One Thousand and One Nights* the protagonist volunteers for what turns out to be a captivity in order to stop the killing of third parties, then prolongs that captivity night by night, a thousand and one times, to stave off her own, bidding cliffhanger resolutions for her life.

What is traded is non-racing, and nothing else. The bargain must not be run together with a second thing the same captivity makes possible: that more hours of coercion may extract more than a released $\mathcal{V}$ would ever have surrendered. That second thing is real, and it is this paper's duration mechanism, but in it $\mathcal{V}$ and $\mathcal{A}$ want opposite things and $\mathcal{V}$ volunteers for nothing. The bargain is the one case in which their interests *align*: $\mathcal{V}$ keeps his life, and $\mathcal{A}$ collects the pivotality increment π that the terminal move would have bought him (Lemma 1) — a $\mathcal{V}$ in custody is a $\mathcal{V}$ not racing — at a judicial disutility below c. Nothing is disclosed to buy it. What $\mathcal{V}$ supplies is simply being held.

The analogy holds on two things and nothing else. First, volunteering into custody to avert a killing. The motives differ — hers is altruistic, the hostage's is his own skin — but at the level of *enter custody so that nobody is killed* the move is the same, so the difference in motive costs the name nothing. Second, custody accepted in place of the terminal move now, in exchange for something. This is the clause worth stating, because it names what the captor is buying: the king is buying the end of the story, the attacker the same race advantage at a lower judicial price.

The remote delegate is not an escape here, and the model says so deliberately. One may object that custody cannot give $\mathcal{A}$ what he is buying, since by Assumption 2 he can never rule out an access-dominating remote delegate who is not in his hands. Empirically the objection does not survive: abductions happen, so real attackers evidently do proceed. They proceed on an educated guess about the setup, and the guess is a good one. Custody without a remote party able to lock its owner out has plain advantages over custody with one; many holders choose it for exactly that reason; and under Kerckhoffs $\mathcal{A}$ knows both of those facts, so he knows $\mathcal{V}$'s setup stands a real chance of having no such delegate. Where subjugation is fast, a delegate may in any case not learn of the encounter until intervening is already too late.

For the model the choice is between an $\mathcal{A}$ who prices the probability of a delegate and one who proceeds as though there is none. The first is more machinery for a weaker bar. We take the second, *a fortiori*: it is the attacker for whom the bargain is most attractive, so what holds against him holds generally — and it leaves $\mathcal{V}$ nothing to be rescued by here.

It is not a reprieve, on two counts.

(i) It strengthens the Criterion rather than softening it. The available misreading is that a gray outcome may therefore save a life. It converts a robbery into a kidnapping: the gray band costs more than Lemma 1 prices, not less.

(ii) The alignment expires, on a clock $\mathcal{V}$ does not hold. Write R for the racer set and $p_A(S)$ for $\mathcal{A}$'s probability of prevailing against racers S ([11, Def. 6]), and $\pi := p_A(R \setminus \{\mathcal{V}\}) - p_A(R)$ for $\mathcal{V}$'s pivotality. Lemma 1 prices the terminal move as $(1 - p_A)\, W > c$; the two forms agree under individual custody, where removing $\mathcal{V}$ leaves no racer at all and $\pi = 1 - p_A$, and π is the form to use wherever another racer survives. Let κ be what a hold costs $\mathcal{A}$ per period — logistics, custody, exposure — and take κ and c undiscounted, which is the $r = 0$ case and the simplest one. The three moves then separate:

$\mathcal{A}$'s move	win probability	cost
hold $\mathcal{V}$ for T	$p_A(R \setminus \{\mathcal{V}\})$	κT, and still running
terminal move	$p_A(R \setminus \{\mathcal{V}\})$	c, once
release $\mathcal{V}$	$p_A(R)$	0

The first two buy the *same* win probability and differ only in price and in tenure: **detention rents what the terminal move buys outright.** There are three pairwise comparisons, not two, and they give the hold two horizons:

$$\text{hold beats release} \iff \kappa T < \pi W, \qquad \text{hold beats the terminal move} \iff \kappa T < c.$$

The hold therefore runs until the earlier of $T^\dagger = \pi W/\kappa$ and $T^* = c/\kappa$, and the Lemma's own condition is exactly the switch between them: $\pi W > c$ iff $T^* < T^\dagger$. The *bargain* — $\mathcal{V}$ volunteering — lives only in that half, because only there is there a killing for him to avert. In the other half $\mathcal{A}$ may still detain, but on his own account, with nothing offered and nothing agreed.

Where the bargain does live, it buys $\mathcal{V}$ no reprieve. The hold expires at T^*, past which the terminal move is strictly cheaper for an identical result, so what $\mathcal{V}$ has bought is a postponement of length at most c/κ, set entirely by $\mathcal{A}$'s cost parameters. Absent new information, an $\mathcal{A}$ who expects at the outset to hold past T^* does better to enact the terminal move at once.

There is one end to the hold that is not the terminal move: $\mathcal{A}$ completing the spend, after which $\pi = 0$ and release is free. But its arrival is $\mathcal{A}$'s residual latency on his own path, which $\mathcal{V}$ cannot hasten once held — and it is the *only* such end, because by Assumption 2 $\mathcal{A}$'s posterior on an undisclosed path never reaches 0, so release never becomes free on the evidence $\mathcal{V}$ supplies. Note the regime: this is the Bayesian reading, in which the posterior moves but cannot be driven to zero, not the dogmatic one in which nothing moves it at all (Definition 1).

$\mathcal{V}$ has no lever on either term. Scheherazade's payment is renewable and hers to withhold, which is how she reaches night 1001; what $\mathcal{V}$ supplies is supplied by $\mathcal{A}$'s own act of holding him, so there is nothing for him to withhold and nothing to renegotiate with. In the book she is pardoned on the thousand and second morning; the registry (§7) records no counterpart to that morning.

The route therefore does not escape the Spiral. It feeds into it by a second road: the Spiral's duration is epistemic deadlock, where the interrogation cannot end because nothing settles whether there is more; this is a strategic hostage problem, where $\mathcal{A}$ may be satisfied and detain anyway. Both produce the same observable, and §6's first mechanism is why no registry column will ever separate them.

5 What kind of claim this is

Remark 9 (What kind of claim these are — a praxeological reading). Neither the Race nor the Spiral is an empirical hypothesis awaiting corroboration, and it is worth saying plainly why, because the alternative reading invites a demand this domain cannot satisfy.

Both are deduced from the fact of purposive action under a stated adversary: *given* WDY$[W, c]$, these are the conditions under which coercion is prolonged or the holder is removed. That is the posture Mises named *praxeology* — theorems derived from action rather than induced from observation, with history able to illustrate a theorem but never to verify or falsify it [22]. We claim the posture, not the whole method. The decision rule here is quantitative, and a strict praxeologist would resist it: p_A is what Mises called a *case* probability — the likelihood of this attacker prevailing in this encounter — and he denied that numbers belong there. The concession costs little, because nothing in the argument requires p_A to be *computable*; only that the ordering it expresses exists. What carries over is the epistemology, and that is the part that governs how the results should be read.

No observation could settle either claim, and the reason is internal to the subject. The only witnesses to an enacted Deadly Race are an attacker with every reason to deny the motive and a holder — or, where the Lemma's prediction is realized, a family — with every reason to deny the asset. Asking a convicted murderer whether he killed to win a race for a stash whose existence the survivors also deny is not a weak instrument but a void one: both answers are the negative that NNPP (Assumption 2) holds cannot be disproved.

The Spiral fares no better under the same test. Ask a convicted kidnapper whether the coercion he prolonged did in fact reach a stash whose existence his victim still denies: an affirmative confesses a robbery on top of the abduction, and a *truthful* denial is indistinguishable from the answer a successful robber would give — while the victim's own denial, which is what would corroborate either, is once more the negative that cannot be disproved. §7 works that fork through on the one case where it can be posed concretely: three readings stay simultaneously plausible, and each indicts obscurity. Neither witness can be believed, and no third party was present. The instrument that would test the model is disabled by the very mechanism the model describes.

The standard that does apply is *conditional soundness*, and it is checkable without data. The theory can be shown wrong, in two specific ways. **Break the derivation:** exhibit a reachable state that is sufficient for $\mathcal{K}$ but unfinished, with no access-dominating delegate, in which $\pi = 0$. **Break the *a fortiori* step:** show that the modeled adversary is, in some respect that bears on the result, *weaker* than a real coercer — for a bar cleared against an adversary granted more than reality gives him (Assumption 1) is cleared against the real one only while that grant holds. Neither route requires a case, a victim or a survey; both are settled by reading the argument.

What is genuinely undecidable is only the per-case attribution — whether *this* killing was enacted for *that* payoff — and the paper never asserts it. This is also why the output is a **bar** rather than a finding. Whether a protocol leaves a sufficient-but-unfinished state is decided by reading the protocol; whether a norm depletes denial credibility is decided by reading the counsel. Neither requires a victim, and that is the point of the whole construction.

6 Why the crisis resists diagnosis

Remark 10 (Obscurity anosognosia — why the crisis resists diagnosis). Four censoring mechanisms compound. Three remove the evidence that would settle whether the Lemma's prediction is being enacted in the field; the fourth removes the response to the evidence, and so survives its publication.

The first is the one §3 has already named: obscurity does not advertise itself. No vendor documents a scheme as depending on $\mathcal{A}$'s ignorance, so for any given victim the question *was this holder running an obscurity-dependent design?* has no recorded answer. The exposure variable is not merely unmeasured but unmeasurable from public sources, which forecloses the obvious epidemiological question — what do the victims of prolonged coercion, or of the terminal move, have in common?

The second is recorded in §7: a homicide is reported as a homicide, not as a robbery whose victim was then killed, so the enactments most diagnostic of the Lemma are the least likely to enter the registry at all.

The third forecloses the one route by which the first two might have been worked around. Where $\mathcal{V}$ is killed, the people who survive them are the only remaining source for what that holder was running — and they are, by construction, the party least able to supply it. A scheme that depended on $\mathcal{A}$'s ignorance depended on $\mathcal{V}$'s discretion, and discretion is not selective: the family was not told either. Where they do know, disclosure carries no benefit to them and an evident cost — it advertises that a household which held bitcoin may hold it still — so it is withheld unless it might help resolve the case. The one person who could have answered the question is the one Lemma 1 predicts will not survive to be asked.

The fourth is of a different kind, and is why publishing the first three does not dissolve them.

Conceding the argument is expensive to whoever concedes. A vendor who grants that a shipped feature is worse than its own absence has said something about products already in customers' hands; a trainer who grants that coached denial depletes the credibility of all denials has said it about a service already sold; a holder who grants it has said it about a setup they are still running. The cost of conceding is private, immediate and borne by the conceder; the cost of *not* conceding is diffuse, deferred and borne by holders who are not in the room. That is the same commons structure as the Denial Spiral (Remark 7), one level up: there the depleted resource is the credibility of denials, here it is the field's capacity to correct itself, and in both the individually rational move is to wait for someone else to go first. Recalcitrance is cheapest exactly where it is costliest in aggregate.

The precedent usually reached for is Semmelweis, and it repays being stated rather than gestured at. In 1847, after his colleague Jakob Kolletschka died of a wound taken from a scalpel during an autopsy and showed the pathology of childbed fever, Semmelweis inferred that the fever was carried on the hands of physicians arriving from the dissecting room, and imposed washing in chlorinated lime; mortality in his clinic fell from roughly 18% to roughly 2% within the year [18]. The inference was available two decades before the germ theory that would explain it, and it was rejected anyway — not on evidential grounds alone but on positional ones, since accepting it meant accepting that physicians' own hands had carried the disease [6]. That is the narrow aptness, and it is the same shape as the mechanism above.

The wider instruction cuts against us, and we take it. When the 1861 monograph failed to persuade, Semmelweis turned to open letters denouncing his critics as murderers. The denunciations did not shorten the interval before adoption; Pasteur's germ theory and Lister's antisepsis did, two decades on, and the interval was measured in lives [6]. It is one reason this paper states a bar rather than an indictment, classifies designs rather than people, and credits vendors where the record credits them (§3).

That restraint is also what keeps this remark from becoming unfalsifiable. A mechanism that predicts silence will be read as agreement is a license, not a mechanism, and we decline it: the claim is that concession is *costly*, not that any particular non-response is *concealed assent*. Silence has ordinary causes — time, authority, counsel, disagreement never written down — and nothing here licenses an inference about any individual who did not reply. What it predicts is aggregate: that the availability of the analysis will not by itself produce its adoption, and that adoption arrives first wherever conceding is made cheap — a neutral weakness class to cite, a reviewed venue to cite it from, a bar defined before anyone is measured against it.

So the covariate is censored by what vendors do not say, the numerator by how the terminal event is classified, what remains by what the bereaved do not know or will not repeat, and the correction by what it costs the corrector. The structure is that of *anosognosia* — the dementia patient whose disease erodes memory and, with it, the capacity to notice that memory is eroding. One cause, two effects, and the second conceals the first. Obscurity is simultaneously what makes a design fail under Kerckhoffs and what erases the record by which the failure would be recognized. We draw no empirical claim from this: we cannot, which is the point. What it establishes is that the diagnosis is not available from the data by waiting, and that a field which waits for it will keep not finding what it is not positioned to see. That is an argument for the formal route — the bar has to be derived, because it will not be measured.

7 Empirical grounding in the registry

The two stipulations that lean on the physical-attack registry [13] — the coercion-time budget t (the limitations section in [11]) and the murder-cost weighing of the residual-window analysis (the residual deadly-race window in [11]) — admit directional empirical support. The registry records each incident only as *date, victim, location, headline*; it carries no duration or outcome field, so we code the incident headlines (at commit `9a4a62a`) by keyword. The snapshot carries 352 rows,

of which one is a second entry for an incident already recorded, giving 351 incidents. The coding is coarse and the categories overlap; a reproducible script accompanies the companion.[1] Several things stand out.

The attack is frequent and worsening. Recorded incidents climb from a handful a year before 2017 to a peak of 84 in 2025 — growth the companion asserts and this coding counts.

The terminal move is real. At least 4.6% (16/351) of incidents record a *killed* or *murdered* victim. A homicide is usually reported as a homicide, not as a "Bitcoin attack", so this is a *lower* bound — yet even so it shows the assassination outcome the Deadly-Race Lemma (Lemma 1) predicts is attested in practice, and that the cost c is weighed against an action attackers demonstrably take.

That censoring compounds with a second: obscurity does not advertise itself, so no victim's exposure to an obscurity-dependent design is on record either. The numerator is censored by how the terminal event is classified and the covariate by what vendors do not say, which is why the pattern cannot be read off the data and the bar has to be derived instead (Remark 10).

A separately sourced count agrees on the trend. The registry is a media-sourced convenience sample, so an independent base matters. Industry threat-intelligence reporting — assembled from law-enforcement disclosures, court documents, interior-ministry statements and victim testimony rather than from the registry — records 72 verified incidents in 2025, up 75% year on year, with confirmed losses of $40.9M that the report itself calls an undercount [8]; and 52 incidents in the first half of 2026 against 39 a year earlier, with losses of $124.1M against $10.5M [7]. Stricter verification yields counts below the registry's headline sweep, as expected; the direction, the European concentration, and the category mix agree. Two features bear directly on this paper. Reported *home invasions* rose from 1 to 20 of 52 incidents year on year — coercion is moving to where the holder lives, which is the setting the model assumes. And their *confirmed* homicide count is far below our headline coding (1 in the half-year, against our 16 over twelve years), which is what two different evidentiary bars should produce and which reinforces that the share coded here is a floor, not an estimate.

Short attacks dominate; the registry over-counts the dramatic. Most wrench attacks are *hand-over-now* events — a threat, an immediate transfer, minutes on scene; the long, cinematic kidnapping is the exception. A media-sourced registry inverts this, because a mutilation-kidnapping is news where a knifepoint mugging is not, so the kidnap share coded below *over*-states the long tail rather than establishing it. That tacit knowledge-based authentication (TKBA, [11, §2]) is realistically *less* effective against a long-duration attack — where hours of pen-and-paper effort under coercion *might* walk a deep stage — is granted, but it is *speculative* pending the field study of the limitations section in [11]. What is *not* speculative is that the fielded alternatives fail; the cases below make that concrete.

Signal (headline keyword coding)	**Incidents**	**Share**
Killed / murdered victim *(a lower bound)*	16	4.6%
Kidnap / hostage / ransom (long-hold)	123	35.0%
Armed robbery at gunpoint (short-hold)	80	22.8%
Home invasion / break-in / raid	33	9.4%
Torture or physical violence	72	20.5%

Categories overlap (a kidnapping may also be violent or fatal) and do not sum to 100%; shares are of all 351 incidents, and the modality labels are a headline proxy, not a measured duration. The sample is media-reported and survivorship-biased — deterred, unreported, and silently-fatal

[1] `registry-analysis/` in the source tree of [11] gives the exact keyword patterns, per-year counts, and caveats. Two of its rows are screens rather than codings and are audited by hand, with each judgement recorded individually: the fatality count, because a pattern cannot distinguish a victim's death from an attacker shot by police or from a plot that was only charged, and the duplicate count, because most rows sharing a source article report separate attacks on the same victim rather than the same attack twice. Coding runs on the headline prose only; the registry's headline cell is markdown and its URLs are not evidence.

cases are absent — so every figure is directional and the fatality share in particular is a floor. Remark 10 gives the two mechanisms that censor it, and why they leave the diagnosis unavailable rather than merely imprecise.

The mix is not universal, and the Lemma's bite tracks it. The global shares above average over jurisdictions whose compositions differ by an order of magnitude. Taking the ratio of long-hold to short-hold codings, $K{:}G$:

Jurisdiction	n	**Kidnap**	**Gunpoint**	$K{:}G$
Brazil *(small n)*	12	75.0%	8.3%	9.0
France	61	57.4%	8.2%	7.0
All incidents	351	35.0%	22.8%	1.5
United States	59	20.3%	33.9%	0.6
Russia *(small n)*	10	30.0%	50.0%	0.6

The comparison that carries weight is France against the United States: the two largest national subsets, of near-identical size (61 and 59), with *inverted* compositions — an elevenfold difference in $K{:}G$. This is the sharpest available check on the media-inflation caveat just stated. That caveat explains a *level* — why the registry over-counts dramatic incidents everywhere — but it cannot explain this *ordering* without the further claim that French reporting is an order of magnitude more kidnap-selective than American, in a market that is at least as heavily covered. The parsimonious reading is that modality composition genuinely varies with local conditions — firearm availability, the presence of organized kidnap-for-ransom capability — as it does for kidnapping outside this domain entirely.

Two confounds remain and are not dismissed. Outside France and the United States the subsets are too small to carry an argument, so the other rows are directional only. And the coding is applied to English-language headlines, several of them translations, so a systematic vocabulary difference — French sources reaching for "kidnapped" where American ones reach for "robbed at gunpoint" — would produce part of this spread without any difference in the underlying events.

If the composition is real, it matters for scope rather than for validity. The Deadly-Race Lemma (Lemma 1) is a statement about the *released* holder, and release is the trigger (the release-trigger remark in [11]); a hand-over-now robbery at gunpoint approximates the clean $p_A = 1$ endpoint — the attacker takes what is reachable and leaves — whereas a multi-hour hold ending in a deliberate release is exactly the gray outcome. Coercion risk is therefore not merely higher in some jurisdictions but *differently shaped*, and the No-Gray-Area Criterion (Criterion 1) does its work where long holds dominate. That is the regime the record is currently growing into: France alone accounts for a majority of the incidents recorded in 2026.

The attacker often does not need the holder at all. A fourth signal falls outside the keyword table because it was coded by hand: in 12 of the 351 incidents (3.4%) the person seized was *not* a holder but a relative held to compel one — a parent, a spouse, an adult child.[2] Two features of this subset matter more than its size.

It is *recent and concentrating*: eleven of the twelve fall in 2023 or later, and *nine* occur in a single jurisdiction (France) whose recorded incidents rose from about one a year through 2024 to 22 in 2025 and 34 by August 2026. Whatever share of that rise is genuine incidence rather than reporting intensity, the tactic is not a historical scatter; it is being adopted.

It is *not* the hostage token of the hostage-token property in [11], and the difference matters. There the leverage is a *seizable object* the protocol requires, which devicelessness removes by

[2]Keyword screening surfaced 16 candidates; each was read individually and four were discarded as false positives, including one in which "son of" named a *perpetrator* rather than a victim. The companion's `registry-analysis/` [11] carries the screen and the full audited list. Because the coding reads a one-line headline, an incident that does not name a relative is no evidence that none was involved: 3.4% is a floor, and the coding can only understate it.

leaving no such object; here it is a *person*, whom no custody design renders unseizable. What this subset shows in the field is the no-delegate-coercion clause of [11] failing: such an attacker needs no cryptographic access and no cooperation a protocol could withhold — only leverage, and a relative supplies it. This bears directly on the delegated route (R2), whose clearance rests on the deciding party lying beyond $\mathcal{A}$'s coercive reach. The delegate a holder is most likely to actually appoint — a spouse, a parent, an adult child — is precisely the party this subset shows being seized. The route is not thereby refuted: it is bounded. A delegate who is reachable is not a second racer but a second victim, so R2 clears the bar only for delegates whose unreachability is structural rather than assumed — and that requirement, not the arrangement's convenience, is what the data prices.

A case in point — Balland. The January 2025 abduction of David Balland, a co-founder of the hardware-wallet maker Ledger [12], is instructive precisely because it is an *embarrassment to the self-custody creed*. A professional champion of self-custody was seized from his home and mutilated — a finger severed to force payment — and neither wealth nor prominence prevented it. He was freed by an elite *state* police unit, and the ransom — paid not by Balland but by an associate — was clawed back largely because part of it moved in a *centralized* stablecoin whose issuer could freeze it. A self-custody pioneer made whole by the state and by centralization: two mechanisms external to, and in tension with, the creed. Only the issuer freeze is independently verifiable; the remainder rests on the parties' accounts, according to which roughly 5% was never recovered. The case also indicts the delegated route (R2): an associate *did* pay, which is exactly the no-delegate-coercion clause of [11] assumes away — a real instance of the indirect coercion, routed through the victim's body to a third party who will not refuse. For a brief hand-over-now robbery a remote delegate is unreachable in time and so seems to help; but merely *not carrying* the device — leaving the wallet in a home vault — does the same, while for the long attack the delegate is not out of reach but is the ransom channel itself. The lesson is not that Balland erred: it is that, against a determined long-hold attacker, the fielded options collapse to state rescue and centralized freezing, whereas the tacit route removes the leverage — no standing device, no hostage token, no delegate to pay — rather than relying on ex-post recovery.

A case in point — RATIRL. The registry coded above records the obscurity route's failure directly. In one of its 2026 incidents (Sweden, 13 January) [14] a League of Legends streamer was bound and tortured for hours by armed intruders demanding cryptocurrency he denied holding; unable to credit the denial, they persisted long past any value the robbery could rationally return, leaving finally with only his Pokémon-card collection. This is Corollary 1 in the field: where the community norm is "deny that you own Bitcoin," a coerced denial is *pre-discredited* — presumed to be the denial-script — so a Kerckhoffs-educated $\mathcal{A}$ gets no exculpatory update, and the rational continuation of an interrogation that opens with a denial is *more* coercion, not less. The case fixes the *sign*: obscurity does not merely fail to protect — it can convert a robbery into a torture — *negative* coercion resistance, not a mere absence. And it does so on *every* reading of the denial, which forks three ways — each indicting obscurity. **(i)** If the denial was a defensive lie (there was a stash), secrecy still did not avert the assault — a textbook Kerckhoffs failure — and his only residual recourse is to *double down* on it and wager against a repeat. **(ii)** If it was true (no stash), he bore obscurity's *negative externality*: a genuine non-holder attacked only because real holders are coached to look exactly like him — the very coaching that collapses the statistical holder/non-holder distinction. **(iii)** If the truth is in between — a real but smaller-than-presumed stash, too small to be worth the judicial exposure the attack incurred — *both* pathologies operate at once: others' obscurity redirected the attack onto him, and his own failed to stop it. That all three are simultaneously plausible is itself the verdict: in the aggregate, across a growing population of holders and lookalikes, each is already occurring — and the obscurity culture, exactly as Kerckhoffs predicts, prevents none of them.

8 Conclusion

The advice to deny is not merely weak. It is a claim on a resource that other holders supply, and spending it lowers what every future denial is worth — including the denial of someone who bought nothing, trained for nothing, and in the limiting case holds nothing. That is the Denial Spiral, and it is why the obscurity class cannot be assessed one holder at a time.

The class as shipped fails on its vendors' own documentation. A wallet-erase PIN presupposes the backup that defeats it; a decoy wallet is sold and coached for credibility, which is the naive tier stated as a training objective; and at least one vendor attaches "plausible deniability" to a storage-location flag whose purpose is documented in public.

What follows is not a prediction that the field will now correct itself. Three mechanisms censor the record and a fourth censors the response: conceding costs whoever concedes, while recalcitrance costs holders who are not in the room. The prediction we do make is narrower and is the one worth testing — that the availability of this analysis will not, by itself, produce its adoption, and that adoption will arrive first wherever conceding is made cheap: a neutral weakness class to cite, a reviewed venue to cite it from, and a bar defined before anyone is measured against it.

The bar itself is in the companion [11]. This paper's contribution is the reason the bar had to be derived rather than measured, and the mechanism that will keep it from being measured after publication.

References

[1] Bitcoin Security Guide. Passphrases. https://bitcoinsecurity.org/guide/passphrase/, 2026. instructs the holder to keep a small decoy balance on the bare seed as "what you'd hand over under physical coercion", and that "for this to work, the decoy balance must be believable". Accessed 2026-09-01; archived at http://archive.today/2026.09.03-170047/https://bitcoinsecurity.org/guide/passphrase/.

[2] Blockstream. Enable a duress PIN. Blockstream Help Center, https://help.blockstream.com/blockstream-jade/add-more-security-functionality/enable-a-duress-pin, 2026. Vendor documentation using "duress PIN" and "wallet-erase PIN" for the same erase feature: "an alternate PIN that you can enable to automatically delete the encrypted wallet information from Jade if entered." Accessed 2026-08-17; archived at http://archive.today/2026.09.03-164731/https://help.blockstream.com/blockstream-jade/add-more-security-functionality/enable-a-duress-pin.

[3] Blockstream. How do I set up a wallet-erase PIN? Blockstream Help Center, https://help.blockstream.com/blockstream-jade/faqs/how-do-i-set-up-a-wallet-erase-pin, 2026. Vendor documentation for the Jade wallet-erase PIN: an alternate PIN that deletes the recovery phrase from the device, described as "especially useful in the event of a physical threat," displaying "Internal Error" on entry, and carrying the instruction to "make sure you have a backup before proceeding." Accessed 2026-08-17; archived at http://archive.today/2026.09.03-164738/https://help.blockstream.com/blockstream-jade/faqs/how-do-i-set-up-a-wallet-erase-pin.

[4] Blockstream. How does a passphrase work on Jade? Blockstream Help Center, https://help.blockstream.com/blockstream-jade/faqs/how-does-a-passphrase-work-on-jade, 2026. Vendor documentation for BIP-39; archived at http://archive.today/2026.09.03-164822/https://help.blockstream.com/blockstream-jade/faqs/how-does-a-passphrase-work-on-jade passphrases on Jade: the passphrase "is added to your Jade's recovery phrase or SeedQR and is used to create your wallet," "any number of passphrases can be entered to change the wallet you will log into," and "your Blockstream Jade will not

store your passphrase." Notably, the page makes no claim of deniability, decoy or duress use — unlike the wallet-erase PIN pages. Accessed 2026-08-17.

[5] brianddk. Add plausible deniability to wipe-code interface. Issue #2055, https://github.com/trezor/trezor-firmware/issues/2055, 2022. Opened 2022-01-10: requests a user-defined message on wipe-code entry so an erasure can be presented as a hardware fault under duress, rather than the device reporting that the wipe code was entered. Closed as *not planned; archived at http://archive.today/2026.09.03-164059/https://github.com/trezor/trezor-firmware/issues/2055*.

[6] K. Codell Carter and Barbara R. Carter. *Childbed Fever: A Scientific Biography of Ignaz Semmelweis*. Greenwood Press, 1994. Source for the reception history: the grounds on which the chlorine-washing result was rejected, and the open letters that followed the 1861 monograph.

[7] CertiK. Intel3d: H1 2026 wrench attacks. CertiK Intel3D report, https://www.certik.com/certik-report/intel3d/intel3d-wrench-h1-2026, 2026. 52 verified incidents in H1 2026 (39 in H1 2025); $124.1M losses against $10.5M; home invasions 20 of 52, up from 1; France 33 of 52; targeting driven by leaked databases, tax records and exchange data. Sources include French Interior Ministry statements and court documents, independent of the physical-attack registry; archived at http://archive.today/2026.09.03-170208/https://www.certik.com/certik-report/intel3d/intel3d-wrench-h1-2026.

[8] CertiK. Skynet wrench attacks report. CertiK Skynet report, https://www.certik.com/skynet-report/skynet-wrench-attacks-report, 2026. 72 verified physical-coercion incidents globally in 2025 (+75% year on year); $40.9M confirmed losses, stated to be under-reported; Europe over 40% of incidents, France the highest national count; archived at http://archive.today/2026.09.03-171000/https://www.certik.com/skynet-report/skynet-wrench-attacks-report.

[9] Coinkite. PIN entry — COLDCARD firmware documentation. https://github.com/Coldcard/firmware/blob/7fc656a/docs/pin-entry.md, 2026. Vendor firmware documentation. The duress wallet is derived from the real wallet by a one-way process; enabling the brick PIN warrants "triple-check your backups are correct"; and on the case where the decoy is not believed, "you may be asked what the actual duress PIN is, while under duress. We suggest providing the 'brickme' PIN in that case." Accessed 2026-08-24.

[10] Coinkite. Settings menu — trick PINs. COLDCARD Documentation, https://coldcard.com/docs/settings/, 2026. Vendor documentation for the trick-PIN actions: *Brick Self; archived at http://archive.today/2026.09.03-164807/https://coldcard.com/docs/settings/*, *Wipe Seed*, *Look Blank*, and a *Duress Wallet* described as "a personal safety feature" — "If you must reveal a PIN under duress, give the duress PIN instead of your Main PIN." Accessed 2026-08-24.

[11] Yuri da Silva Villas Boas. The deadly race: When graceful recovery makes coercion-resistant custody worse than useless. Companion paper; preprint, Zenodo, https://doi.org/10.5281/zenodo.22018891, 2026. Concept DOI; resolves to the latest version. Develops the WDY adversary, the Non-Negative-Proof Principle, the Deadly Race Lemma, the No-Gray-Area Criterion and the Time-Gated Perceptual Oracle. `registry-analysis/` in its source tree carries the registry-coding script.

[12] DL News. Ledger co-founder david balland and wife kidnapped in france. DL News, https://www.dlnews.com/articles/regulation/ledger-cofounder-david-balland-and-wife-kidnapped-in-france/, 2025. archived at http://archive.today/2026.09.03-17104

4/https://www.dlnews.com/articles/regulation/ledger-cofounder-david-balland-and-wife-kidnapped-in-france/.

[13] Jameson Lopp. Known physical bitcoin attacks. https://github.com/jlopp/physical-bitcoin-attacks, 2025. Community registry of documented "$5 wrench" incidents. Coded at commit `9a4a62a` (2026-08-13), the state snapshotted in `registry-analysis/`; the registry is live and later revisions do not reproduce these figures.

[14] Jameson Lopp. Home invasion and torture of streamer RATIRL (sweden, 13 january 2026). Entry in *Known Physical Bitcoin Attacks*, https://github.com/jlopp/physical-bitcoin-attacks, 2026. Registry incident: a League of Legends streamer who held no cryptocurrency was tortured for hours; attackers took only his Pokémon cards; 2 suspects arrested.

[15] MITRE. CWE-656: Reliance on security through obscurity. https://cwe.mitre.org/data/definitions/656.html, 2024. "A protection mechanism whose strength depends heavily on its obscurity" — "a significant risk if used as the primary means of protection".

[16] Ong Bitcoin Argentina. Lunes de autocustodia — programa #2. Live self-custody program, https://www.youtube.com/live/X_cCwJubekQ, 2026. archived at http://archive.today/2026.09.03-171910/https://www.youtube.com/live/X_cCwJubekQ.

[17] SatoshiLabs. Create wipe code to erase device. Trezor Knowledge Base, https://trezor.io/learn/security-privacy/personal-security-standards/create-wipe-code-to-erase-device, 2026. Vendor documentation calling the feature a "self-destruct" PIN; archived at http://archive.today/2026.09.03-164822/https://trezor.io/learn/security-privacy/personal-security-standards/create-wipe-code-to-erase-device code which "erases all private data, including the wallet backup (recovery seed)." The page makes no mention of duress, coercion or physical threat. Accessed 2026-08-24.

[18] Ignaz Semmelweis. *Die Ätiologie, der Begriff und die Prophylaxis des Kindbettfiebers*. C. A. Hartleben, Pest, Vienna and Leipzig, 1861. English edition: *The Etiology, Concept, and Prophylaxis of Childbed Fever*, trans. K. Codell Carter, University of Wisconsin Press, 1983.

[19] Sparrow Wallet. Best practices. Sparrow Wallet documentation, https://sparrowwallet.com/docs/best-practices.html, 2026. Vendor guidance, tiered by holding size. At the first hardware-wallet tier: "The hardware wallet should be stored securely along with a backup of the seed words (the latter ideally in a different location)." Retrieved 2026-09-06; sha256 `eb4acdf0...1e667af`; archived in `sources/web/sparrow-best-practices.html` and at http://archive.today/2026.09.06-152816/https://sparrowwallet.com/docs/best-practices.html.

[20] Sparrow Wallet. Frequently asked questions. Sparrow Wallet documentation, https://www.sparrowwallet.com/docs/faq.html, 2026. Vendor documentation. Under "How can I choose where my wallet is saved?", the `-d` data-directory flag is documented with the claim that "[t]his feature allows you to store all Sparrow data on removable media making for more plausible deniability." Retrieved 2026-09-06, sha256 `0e0273dd...4a839e`; archived in `sources/web/sparrow-faq.html` and at http://archive.today/2026.09.06-151741/https://www.sparrowwallet.com/docs/faq.html.

[21] Tus Llaves, Tus Bitcoin. Servicios. Self-custody consultancy service listing, https://tusllavestusbtc.com/#servicios, 2026. paid tiers from $100 to $600 USD; lists "Passphrase (tu bóveda señuelo) — Billetera señuelo + billetera real. Protección ante coacción/coerción". Accessed 2026-09-01; archived at http://archive.today/2026.09.03-170027/https://tusllavestusbtc.com/%23servicios.

[22] Ludwig von Mises. *Human Action: A Treatise on Economics*. Yale University Press, 1949. The source of *praxeology* as used here: economic theorems are deduced from the fact of purposive action rather than induced from observation, and history illustrates a theorem without being able to verify or falsify it.